

HW 2 Aidan Bugayong

```
# install.packages('GGally', repos = "http://cran.us.r-project.org")
# install.packages("car", repos = "http://cran.us.r-project.org")
# install.packages("faraway", repos = "http://cran.us.r-project.org")

library(faraway)
library(GGally)
library(car)
```

1

```
data = read.csv("data/slumptest-1.csv", sep=",")

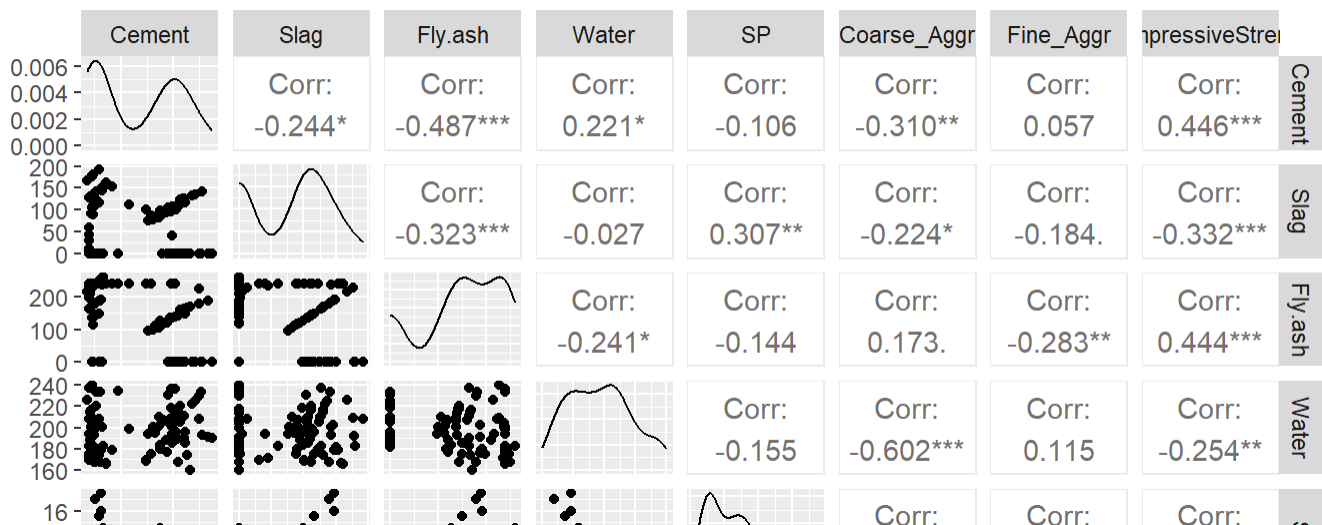
dim(data)
```

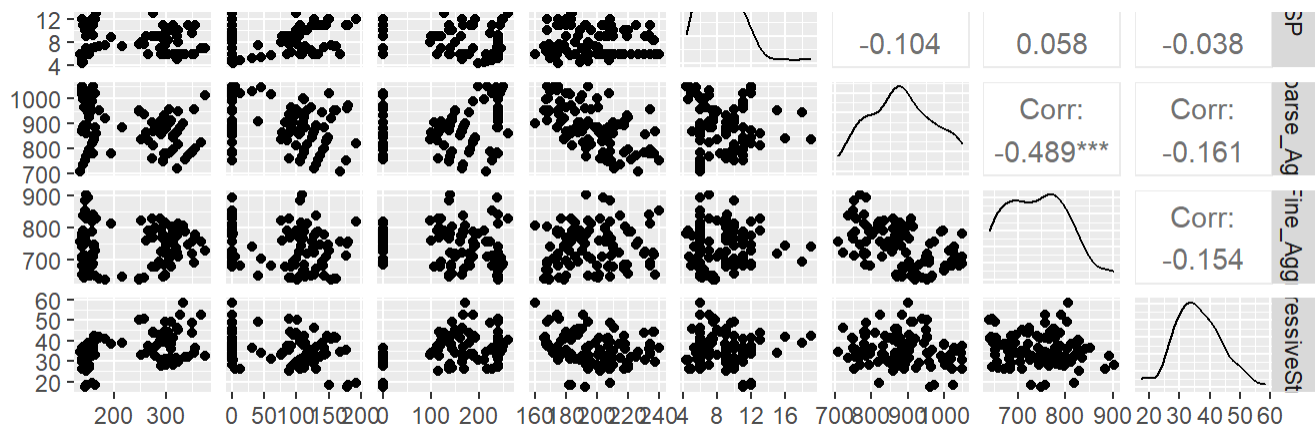
```
[1] 103    8
```

```
head(data)
```

	Cement	Slag	Fly.ash	Water	SP	Coarse_Aggr	Fine_Aggr	CompressiveStrength
1	273	82	105	210	9	904	680	34.99
2	163	149	191	180	12	843	746	41.14
3	162	148	191	179	16	840	743	41.81
4	162	148	190	179	19	838	741	42.08
5	154	112	144	220	10	923	658	26.82
6	147	89	115	202	9	860	829	25.21

```
ggpairs(data)
```





```
fit <- lm(CompressiveStrength ~ ., data=data)
summary(fit)
```

Call:

```
lm(formula = CompressiveStrength ~ ., data = data)
```

Residuals:

Min	1Q	Median	3Q	Max
-5.8411	-1.7063	-0.2831	1.2986	7.9424

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	139.78150	71.10128	1.966	0.05222 .
Cement	0.06141	0.02282	2.691	0.00842 **
Slag	-0.02971	0.03176	-0.935	0.35200
Fly.ash	0.05053	0.02316	2.182	0.03159 *
Water	-0.23270	0.07166	-3.247	0.00161 **
SP	0.10315	0.13459	0.766	0.44532
Coarse_Aggr	-0.05562	0.02744	-2.027	0.04546 *
Fine_Aggr	-0.03908	0.02882	-1.356	0.17833

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 2.609 on 95 degrees of freedom

Multiple R-squared: 0.8968, Adjusted R-squared: 0.8892

F-statistic: 118 on 7 and 95 DF, p-value: < 2.2e-16

i

According to the above summary, coarse aggregate has a p-value of 0.04546 which is less than $\alpha = 0.05$ so we can say that coarse aggregate is a statistically significant predictor.

ii

```
"" ~ ., data = data)
```

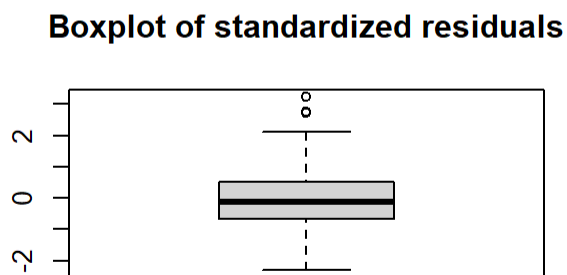
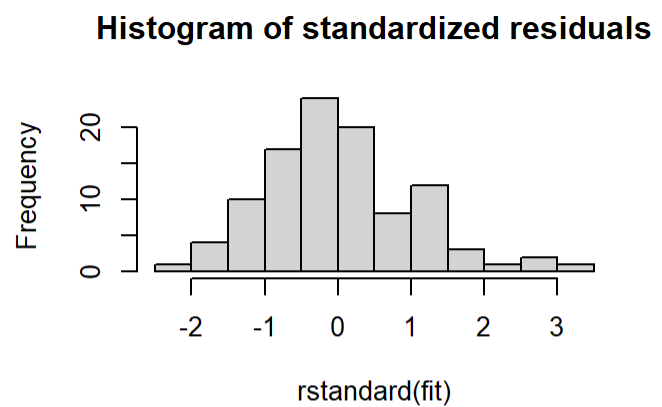
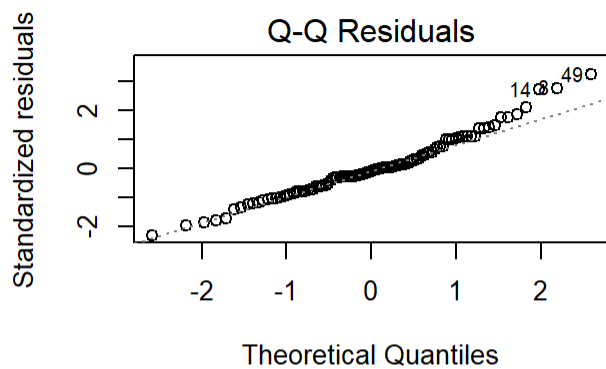
```
## Compute the residuals
std_resid <- fit$residuals

par(mfrow=c(2,2))
plot(fit, which=2)
hist(rstandard(fit), main="Histogram of standardized residuals")
boxplot(rstandard(fit), main="Boxplot of standardized residuals")

## Shapiro-Wilk's test
shapiro.test(std_resid)
```

Shapiro-Wilk normality test

data: std_resid
W = 0.97486, p-value = 0.0467



Here the p-value is 0.0467 which is less than $\alpha = 0.05$ so we reject and thus the data is not normally distributed.

iii

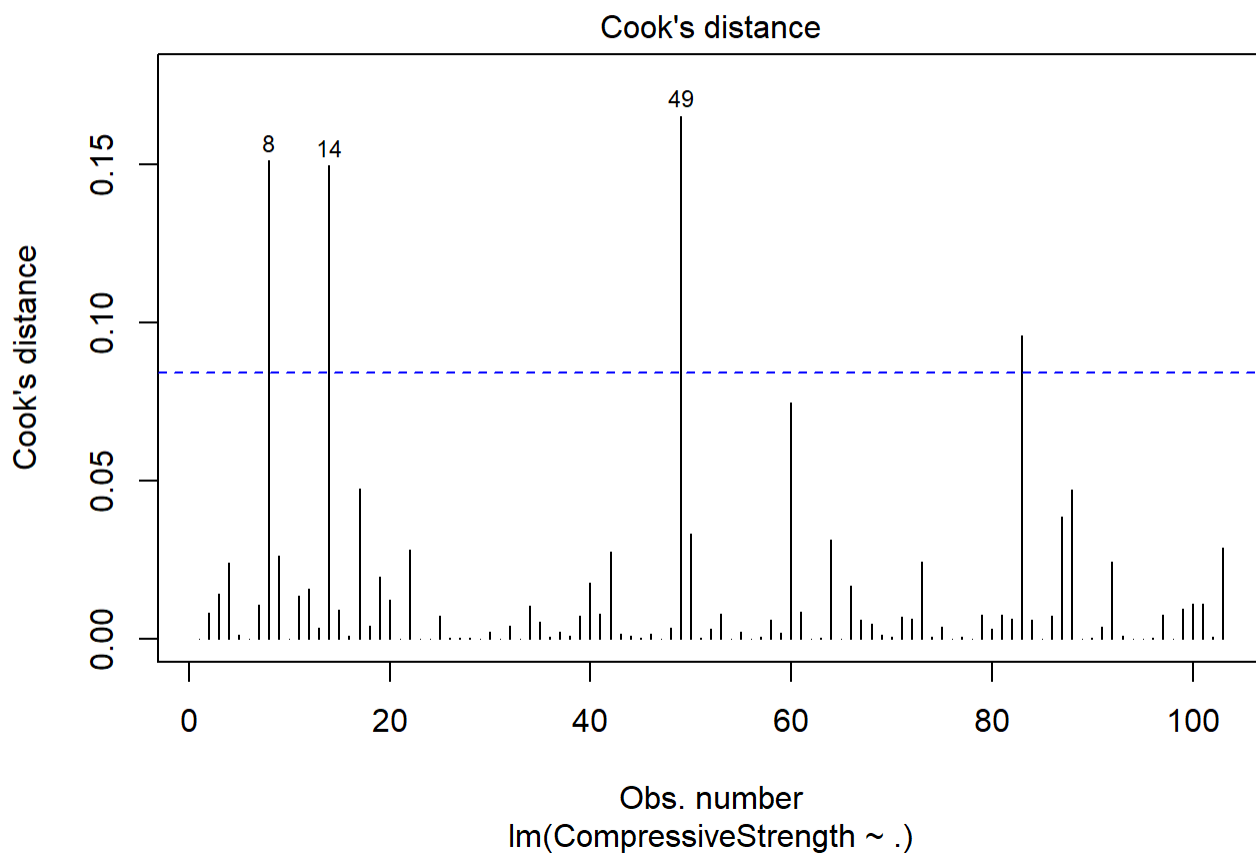
```
## Cook's distance
```

```
D <- cooks.distance(lm(fit))
cutoff1 <- qf(0.5, 8, df.residual(fit))
print(cutoff1)
```

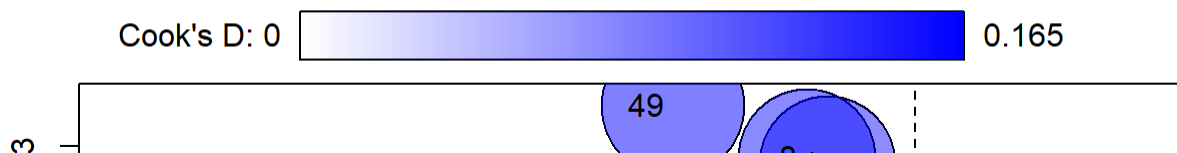
```
[1] 0.9245399
```

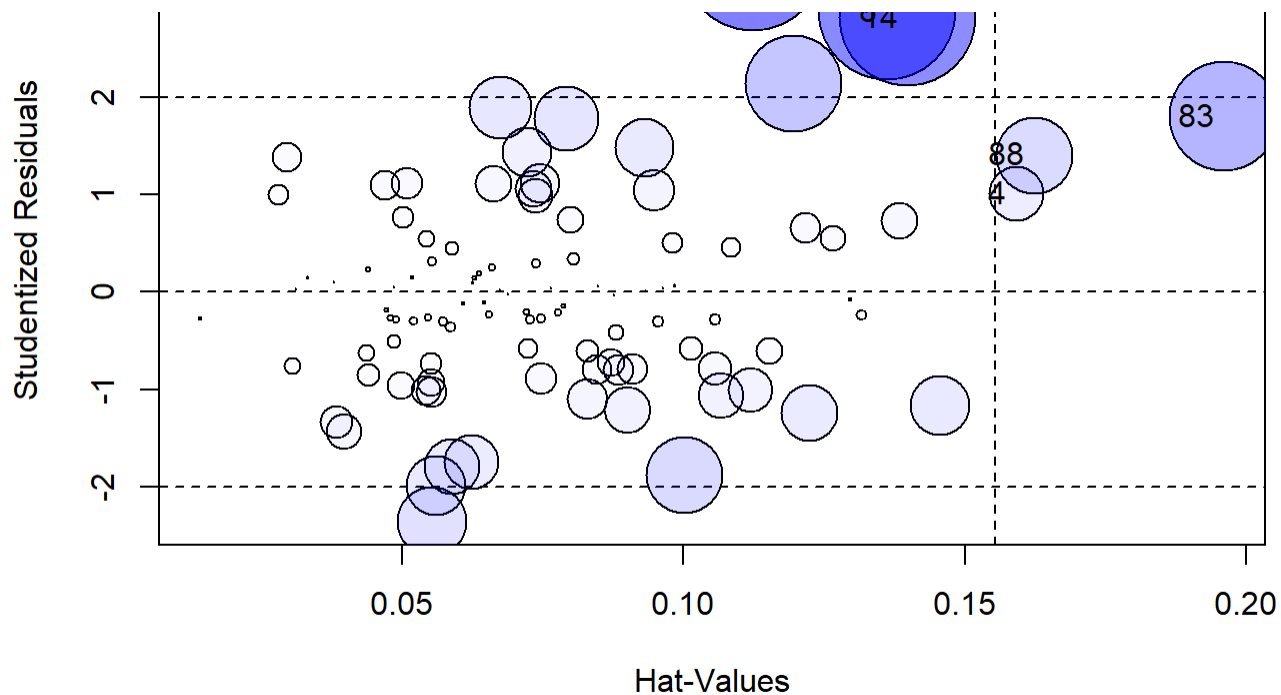
```
cutoff2 <- 8/df.residual(fit)
cutoff3 <- 0.5

plot(fit, which=4)
abline(h=cutoff1, lty=2, col="black")
abline(h=cutoff2, lty=2, col="blue")
abline(h=cutoff3, lty=2, col="red3")
```



```
influencePlot(fit, id=list(n=3))
```





	StudRes	Hat	CookD
4	1.007598	0.1590495	0.02399811
8	2.873893	0.1361820	0.15120617
14	2.811296	0.1397976	0.14967778
49	3.407478	0.1124498	0.16540816
83	1.795464	0.1961051	0.09605162
88	1.403472	0.1623378	0.04723434

There are quite a few outliers in the data. These being entries 8, 14, 49 and they are not influential because their cook distances are below the cutoff 1 of 0.925

iv

```
data_no_outliers <- data[-c(8, 14, 49), ]
fit2 <- lm(CompressiveStrength ~ ., data = data_no_outliers)

summary(fit)
```

Call:

```
lm(formula = CompressiveStrength ~ ., data = data)
```

Residuals:

Min	1Q	Median	3Q	Max
-5.8411	-1.7063	-0.2831	1.2986	7.9424

Coefficients:

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```

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Slag        -0.02971    0.03176   -0.935  0.35200
Fly.ash     0.05053    0.02316    2.182  0.03159 *
Water       -0.23270    0.07166   -3.247  0.00161 **
SP          0.10315    0.13459    0.766  0.44532
Coarse_Aggr -0.05562    0.02744   -2.027  0.04546 *
Fine_Aggr   -0.03908    0.02882   -1.356  0.17833
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

```

Residual standard error: 2.609 on 95 degrees of freedom
Multiple R-squared:  0.8968,    Adjusted R-squared:  0.8892
F-statistic: 118 on 7 and 95 DF,  p-value: < 2.2e-16

```

```
summary(fit2)
```

```

Call:
lm(formula = CompressiveStrength ~ ., data = data_no_outliers)

```

```

Residuals:
    Min       1Q   Median       3Q      Max
-5.8661 -1.3657 -0.1508  1.3244  5.4743

```

```

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) 103.520083  62.382326   1.659  0.100431
Cement       0.073248   0.020055   3.652  0.000432 ***
Slag        -0.008788   0.027953  -0.314  0.753945
Fly.ash     0.063659   0.020323   3.132  0.002326 **
Water       -0.200126   0.063029  -3.175  0.002038 **
SP          0.198984   0.118788   1.675  0.097307 .
Coarse_Aggr -0.041415   0.024103  -1.718  0.089112 .
Fine_Aggr   -0.025668   0.025241  -1.017  0.311858

```

```

---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

```

Residual standard error: 2.274 on 92 degrees of freedom
Multiple R-squared:  0.9173,    Adjusted R-squared:  0.911
F-statistic: 145.7 on 7 and 92 DF,  p-value: < 2.2e-16

```

First there is a vast decrease in the intercept from fit1 to fit2. This was compensated by all the other coefficients increasing by 0.01 – 0.02. More importantly are the p-values where cement and fly.ash both gained a * (indicating an increase in statistical significance) while coarse aggregate actually decreased in significance.

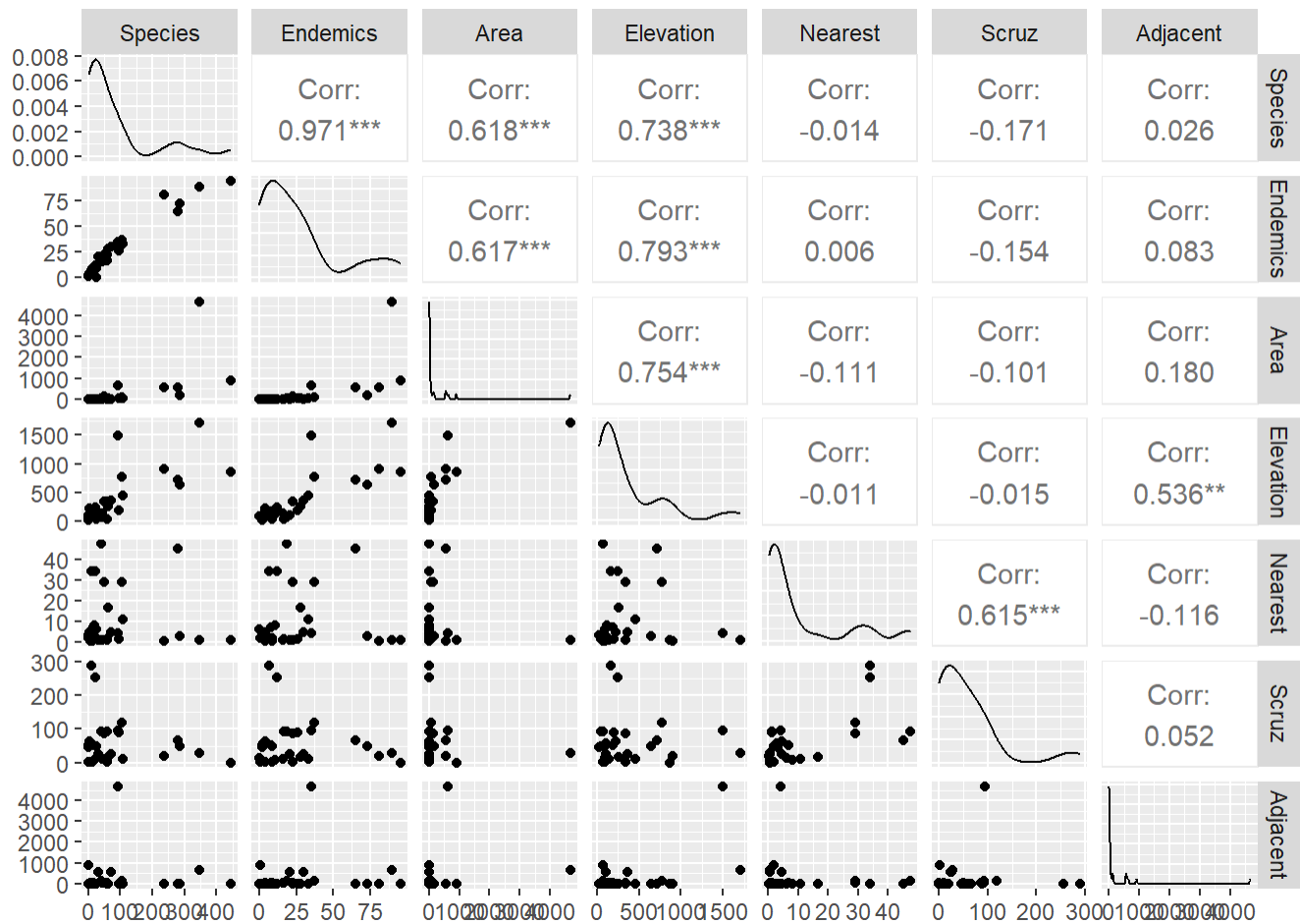
```
data(gala)
dim(gala)
```

```
[1] 30 7
```

```
head(gala)
```

	Species	Endemics	Area	Elevation	Nearest	Scruz	Adjacent
Baltra	58	23	25.09	346	0.6	0.6	1.84
Bartolome	31	21	1.24	109	0.6	26.3	572.33
Caldwell	3	3	0.21	114	2.8	58.7	0.78
Champion	25	9	0.10	46	1.9	47.4	0.18
Coamano	2	1	0.05	77	1.9	1.9	903.82
Daphne.Major	18	11	0.34	119	8.0	8.0	1.84

```
ggpairs(gala)
```



```
fit3 = lm(Species ~ ., data = gala)
```

```
summary(fit3)
```

Call:

```
lm(formula = Species ~ ., data = gala)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-68.219	-10.225	1.830	9.557	71.090

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-15.337942	9.423550	-1.628	0.117
Endemics	4.393654	0.481203	9.131	4.13e-09 ***
Area	0.013258	0.011403	1.163	0.257
Elevation	-0.047537	0.047596	-0.999	0.328
Nearest	-0.101460	0.500871	-0.203	0.841
Scruz	0.008256	0.105884	0.078	0.939
Adjacent	0.001811	0.011879	0.152	0.880

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 28.96 on 23 degrees of freedom

Multiple R-squared: 0.9494, Adjusted R-squared: 0.9362

F-statistic: 71.88 on 6 and 23 DF, p-value: 9.674e-14

i

```
null_fit = lm(Species ~ 1, data = gala)
anova(fit3, null_fit, test="F")
```

Analysis of Variance Table

Model 1: Species ~ Endemics + Area + Elevation + Nearest + Scruz + Adjacent

Model 2: Species ~ 1

	Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
1	23	19295				
2	29	381081	-6	-361787	71.877	9.674e-14 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

We notice from the anova table that the p-value is much smaller than $\alpha = 0.05$ thus our model is useful.

ii

From the summary we note that the adjusted R^2 is 0.9362 which means that 93.62 of the variance of Species is explained by our predictors.

iii

```
vif(fit3)
```

Endemics	Area	Elevation	Nearest	Scruz	Adjacent
5.979425	3.356530	13.920084	1.767133	1.793814	3.645582

Elevation has a vif above 10 so there is reason to believe multicollinearity is present.

iv

```
reduced_model <- lm(Species ~ Area + Elevation, data = gala)
anova(reduced_model, fit3)
```

Analysis of Variance Table

Model 1: Species ~ Area + Elevation

Model 2: Species ~ Endemics + Area + Elevation + Nearest + Scruz + Adjacent

	Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
1	27	169947				
2	23	19295	4	150652	44.896	1.524e-10 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

The claim is false because the p-value is less than $\alpha = 0.01$ so we reject the null hypothesis thus the predictors are useful.

V

```
ncvTest(reduced_model)
```

Non-constant Variance Score Test

Variance formula: ~ fitted.values

Chisquare = 21.09394, Df = 1, p = 4.3731e-06

Since the p-value is less than $\alpha = 0.05$ the constant variance assumption is violated.